Support vector machine kernel selection

SAheart cardiovascular-risk data: model chd (coronary heart disease; 0=no, 1=yes) from sbp (systolic blood pressure), tobacco, ldl, adiposity, famhist, typea score, obesity, alcohol, age.

n=462; counts 0:302, 1:160; famhist Absent=1, Present=2. Complete data; suitable for binary SVM.

Linear baseline; polynomial degrees 2-4 allow interactions; RBF allows flexible nonlinear boundaries.

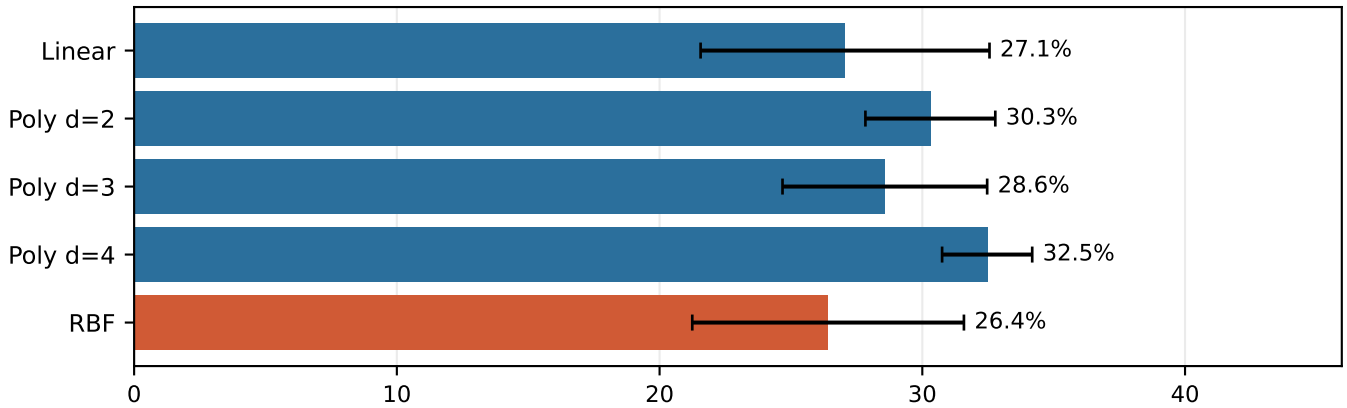
Tune C={0.1, 1, 10}; polynomial gamma=1/9, coef0=0; RBF gamma={0.01, 0.1}.

Shared stratified five-fold CV (seed 34235); scale using each fold's training data only.
Select the lowest mean fold misclassification rate; ties favor simpler kernels, then lower C/gamma.

All 18 settings: mean five-fold CV misclassification rate

Kernel	Tuning settings	Mean CV rate (%)
rbf*	C=10, gamma=0.01	26.41%
rbf	C=1, gamma=0.01	26.62%
linear	C=10	27.06%
linear	C=0.1	27.27%
linear	C=1	27.27%
poly-3	C=1, gamma=1/9, coef0=0	28.57%
rbf	C=1, gamma=0.1	29.23%
poly-2	C=10, gamma=1/9, coef0=0	30.31%
poly-4	C=0.1, gamma=1/9, coef0=0	32.47%
poly-4	C=1, gamma=1/9, coef0=0	32.48%
poly-2	C=1, gamma=1/9, coef0=0	32.90%
poly-3	C=10, gamma=1/9, coef0=0	32.91%
poly-3	C=0.1, gamma=1/9, coef0=0	33.13%
poly-2	C=0.1, gamma=1/9, coef0=0	33.77%
rbf	C=10, gamma=0.1	34.41%
rbf	C=0.1, gamma=0.1	34.42%
poly-4	C=10, gamma=1/9, coef0=0	34.43%
rbf	C=0.1, gamma=0.01	34.63%

* selected candidate



Mean CV misclassification rate (%); error bars show fold SD; lower is better; selected family is orange.

Preferred: RBF, C=10, gamma=0.01 has the lowest mean CV misclassification rate (26.41%).
Next best: RBF, C=1, gamma=0.01 (26.62%); the 0.21-point gap is small relative to fold variation.

Full-data refit: 280 support vectors. CV only (no test set); class imbalance and fold variation limit precision.